

CTMT Morphism Resolution by Automorphisms of Resolution Geometry:

A Conservative Axiom, Theorem Build-Up, Classification Result, and
Sensor-to-Morphism Protocol

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Abstract

This paper gives the current conservative formulation of the CTMT local morphism problem. Earlier formulations tried to pass through a tuning law, Fisher metric, or scalar diagnostic before reaching geometry. The present formulation reverses the abstraction order: observations induce a resolution geometry, admissible morphisms are automorphisms of that geometry, and Fisher/null/monodromy/winding diagnostics are derived consequences.

The primitive object is defined coordinate-free as an observation space decomposed into resolved and unresolved sectors together with the covariance structure induced by the measurement protocol. In a chosen representation this is written

$$\mathcal{G} = (\mathcal{O}; R, N, \Sigma), \quad R = \text{Im } J, \quad N = \text{Ker } J^T,$$

where J is the local observable-response Jacobian and Σ is the observation covariance. The core CTMT axiom is

$$\phi \in \text{Mor}_{\text{CTMT}} \iff \phi^* \mathcal{G} = \mathcal{G},$$

with a conformal replacement $\mathcal{G}_c = (\mathcal{O}; R, N, [\Sigma])$ only when the protocol explicitly discards absolute covariance scale.

The mathematical core is the automorphism classification. In an adapted decomposition $\mathcal{O} = R \oplus N$ with

$$\Sigma = \begin{pmatrix} \Sigma_R & C \\ C^T & \Sigma_N \end{pmatrix},$$

strict automorphisms are precisely block maps $\text{diag}(A, B)$ satisfying

$$A \Sigma_R A^T = \Sigma_R, \quad B \Sigma_N B^T = \Sigma_N, \quad A C B^T = C.$$

Conformal automorphisms satisfy the same equations up to one common positive scale. If $C = 0$, the groups reduce to

$$\text{Aut}(\mathcal{G}) \cong O(r) \times O(n-r), \quad \text{Aut}(\mathcal{G}_c) \cong \mathbb{R}_+ \times O(r) \times O(n-r).$$

If $C \neq 0$, the group is the stabilizer, respectively conformal stabilizer, of the resolved–null covariance coupling. Controlled batteries on coil, Navier-like, diffusion, reaction–diffusion, and seismic local charts are not used as proof of the axiom; they demonstrate that the axiom produces stable, computable, falsifiable diagnostics across heterogeneous observation systems.

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1 Purpose and claim stratification

The purpose of this paper is to state the CTMT local morphism problem in a form that is mathematically evaluable. The core proposal is not that a particular coil, fluid, turbine, or seismic experiment proves CTMT. The core proposal is that local CTMT admissibility is an automorphism condition for a resolution geometry induced by observation.

The paper separates four layers.

Definition/Axiom layer. CTMT admissible morphisms are defined as maps preserving resolution geometry.

Theorem layer. Rank, resolved/null decomposition, covariance compatibility, Fisher invariance, and automorphism group classification follow under stated assumptions.

Operational layer. Sensor samples determine J , R , N , Σ , and hence the candidate automorphism group.

Experimental layer. Batteries demonstrate computability, stability, and falsifiability of the definition. They do not prove the axiom true.

Remark 1 (Change of abstraction). The working order is no longer

$$\text{observation} \rightarrow \text{tuning law} \rightarrow \text{Fisher} \rightarrow \text{geometry}.$$

It is instead

$$\text{observation} \rightarrow \text{resolution geometry} \rightarrow \text{admissible morphisms} \rightarrow \text{Fisher/null/monodromy/winding diagnostics}.$$

This is the main conceptual stabilization achieved by the recent falsification sequence.

2 Observation systems and resolution geometry

2.1 Operational observation chart

Let $\mathcal{O}$ be a finite-dimensional real observation vector space. A measurement campaign provides samples

$$y_k \in \mathcal{O}, \quad k = 1, \dots, m.$$

The samples may be electromagnetic sensor readings, Navier-window observables, turbine-machine telemetry, diffusion profile samples, or seismic catalogue features after a declared preprocessing pipeline. CTMT does not begin with a spacetime metric. It begins with an observation protocol and the response of observables to admissible perturbations.

Definition 1 (Operational observation chart). An operational chart is a quadruple

$$\mathfrak{C} = (\mathcal{O}, F, \Theta, \Sigma),$$

where:

- (a) $\mathcal{O}$ is the observation space;
- (b) Θ is the finite-dimensional space of admissible perturbation variables;
- (c) $F : \Theta \rightarrow \mathcal{O}$ is the observable-response map;
- (d) Σ is the covariance structure estimated or specified by the measurement protocol.

The covariance structure includes the protocol statement determining whether absolute scale is calibrated or quotiented out.

Definition 2 (Observable-response Jacobian). At a base point $\theta_0 \in \Theta$, the local observable-response Jacobian is

$$J = D_{\theta}F(\theta_0) : T_{\theta_0}\Theta \rightarrow \mathcal{O}. \quad (1)$$

In data applications, J may be estimated by finite differences, local linear regression, automatic differentiation of the reconstruction map, or another declared estimator. Rank stability and conditioning are part of the reported chart quality.

2.2 Resolved and unresolved sectors

Definition 3 (Resolved and unresolved sectors). The resolved sector is the observable first-order response image

$$R := \text{Im } J \subset \mathcal{O}. \quad (2)$$

The unresolved, or null, sector is

$$N := \text{Ker } J^T \subset \mathcal{O}, \quad (3)$$

where J^T is taken with respect to the declared background pairing used to represent sensor coordinates. Equivalently, N consists of observation directions invisible to the chosen perturbation protocol at first order.

Remark 2 (Null does not mean absent). The null sector is present in the observation vector. It is simply not activated by the declared admissible perturbations at first order. Changing it can therefore be operationally relevant even when Fisher pullback diagnostics appear unchanged.

2.3 Coordinate-free resolution geometry

The notation (R, N, Σ) is convenient but should not be read as the final object. The final object is the decomposition and covariance structure on the observation space, independent of the basis used to write matrices.

Definition 4 (Strict resolution geometry). A strict resolution geometry is

$$\boxed{\mathcal{G} = (\mathcal{O}; R, N, \Sigma),} \quad (4)$$

where $R, N \subset \mathcal{O}$ are complementary operational sectors induced by J , and Σ is the calibrated covariance structure on $\mathcal{O}$.

Definition 5 (Conformal resolution geometry). A conformal resolution geometry is

$$\boxed{\mathcal{G}_c = (\mathcal{O}; R, N, [\Sigma]), \quad [\Sigma] = \{\lambda \Sigma : \lambda > 0\}.} \quad (5)$$

It is admissible only when the observation protocol explicitly declares global covariance scale unobservable or normalized away.

Definition 6 (Sector-relative candidate geometry). A sector-relative geometry is a possible weaker structure

$$\mathcal{G}_s = (\mathcal{O}; R, N, \Sigma_R, \Sigma_N, C_{RN} \text{ modulo a declared sector equivalence}), \quad (6)$$

where the equivalence relation may preserve only selected covariance information relative to the decomposition $R \oplus N$. This paper does not adopt (6) as the default CTMT geometry. It is listed as an open protocol option requiring its own falsification battery.

3 Core CTMT axiom

Axiom 7 (Local CTMT morphism axiom). For a fixed operational chart with strict calibrated covariance, a local map $\phi : \mathcal{O} \rightarrow \mathcal{O}$ is an admissible CTMT morphism if and only if it preserves the strict resolution geometry:

$$\boxed{\phi \in \text{Mor}_{\text{CTMT}} \iff \phi^* \mathcal{G} = \mathcal{G}.} \quad (7)$$

If the protocol explicitly identifies covariance only up to global positive scale, the axiom is replaced by

$$\boxed{\phi \in \text{Mor}_{\text{CTMT}} \iff \phi^* \mathcal{G}_c = \mathcal{G}_c.} \quad (8)$$

In a concrete matrix representation, (7) means

$$\phi(R) = R, \quad \phi(N) = N, \quad \phi \Sigma \phi^T = \Sigma. \quad (9)$$

The conformal replacement is

$$\phi(R) = R, \quad \phi(N) = N, \quad \phi \Sigma \phi^T = \lambda \Sigma \quad \text{for some } \lambda > 0. \quad (10)$$

Remark 3 (Axiom, not experimental conclusion). The seismic, coil, Navier, and synthetic batteries do not prove (7). They test whether this axiom produces stable, computable, and falsifiable diagnostics when applied to heterogeneous data-generating systems.

4 Why this geometry is forced by admissibility requirements

This section records the theorem layer that makes $\mathcal{G}$ look less chosen. Each statement is conditional: if CTMT admissibility means preserving the operational identity of a local chart, then the following structures must be preserved.

Proposition 8 (Rank invariance). Let two local observation charts be related by an invertible observation map $\phi : \mathcal{O} \rightarrow \mathcal{O}$ and an invertible reparametrization $a : T_{\theta_0}\Theta \rightarrow T_{\theta'_0}\Theta'$ such that

$$J'a = \phi J. \quad (11)$$

Then

$$\text{rank } J' = \text{rank } J. \quad (12)$$

In particular, any admissible local chart isomorphism preserves the dimension of the resolved sector.

Proof 9. Since a and ϕ are invertible, multiplication by them cannot change matrix rank. From $J'a = \phi J$, we obtain $\text{rank } J' = \text{rank}(J'a) = \text{rank}(\phi J) = \text{rank } J$.

Corollary 10 (Resolved dimension and null dimension are invariants). If $r = \text{rank } J$ and $n = \dim \mathcal{O}$, then

$$\dim R = r, \quad \dim N = n - r$$

are invariants of admissible local chart isomorphisms.

Proposition 11 (Resolved sector functoriality). Under the intertwining relation (11),

$$R' = \text{Im } J' = \phi(R). \quad (13)$$

Thus an endomorphism of one chart is admissible only if $\phi(R) = R$.

Proof 12. Because a is invertible, $\text{Im } J' = \text{Im}(J'a)$. By (11), this equals $\text{Im}(\phi J) = \phi(\text{Im } J) = \phi(R)$.

Proposition 13 (Null sector preservation). Assume the null sector is defined as the annihilator of first-order observable response under the declared observation pairing. If a chart endomorphism preserves the operational distinction between response-visible and response-invisible observation directions, then

$$\phi(N) = N. \quad (14)$$

Equivalently, resolved/null mixing is not an automorphism of the resolution geometry.

Proof 14. By definition, $v \in N$ iff $J^T v = 0$, i.e. iff v has zero first-order response pairing with all admissible perturbations. If ϕ preserves the operational distinction between visible and invisible directions, then v has this property exactly when ϕv has it. Hence $\phi(N) = N$.

Proposition 15 (Covariance compatibility). If two descriptions represent the same calibrated observation protocol, then observational covariance must be preserved:

$$\phi \Sigma \phi^T = \Sigma. \quad (15)$$

If the protocol identifies all global covariance scales as equivalent, then the most general compatible condition is

$$\phi \Sigma \phi^T = \lambda \Sigma, \quad \lambda > 0. \quad (16)$$

Proof 16. The covariance structure determines reproducible observational distinguishability and noise calibration. In a strict protocol, two descriptions of the same measurement must assign the same covariance to transported observations, giving equality. In a scale-quotiented protocol, the only discarded datum is one global positive scale, giving equality up to $\lambda > 0$.

Corollary 17 (Minimal strict invariant). Under the preceding requirements, any admissible local morphism must preserve

$$R, \quad N, \quad \Sigma$$

in the strict case, and

$$R, \quad N, \quad [\Sigma]$$

in the conformal case. Therefore the resolution geometry is the minimal local structure retained by these admissibility requirements.

5 Derived diagnostics

5.1 Fisher geometry

Definition 18 (Fisher recognition metric). When Σ is positive definite, the Fisher pullback metric on perturbation space is

$$G_F = J^T \Sigma^{-1} J. \quad (17)$$

Proposition 19 (Fisher is induced). The Fisher metric is induced by the pair (J, Σ) . It is not a sufficient replacement for the full resolution geometry.

Proof 20. For perturbations $u, v \in T_{\theta_0} \Theta$, covariance-weighted observational distinguishability is

$$\langle Ju, Jv \rangle_{\Sigma^{-1}} = u^T J^T \Sigma^{-1} Jv.$$

Thus G_F is the pullback of observational distinguishability along J . Since G_F only sees the image of J , it can miss changes in the null sector and in covariance components not detected by this pullback.

Proposition 21 (Fisher invariance under strict admissible chart isomorphism). Suppose $J'a = \phi J$ with a invertible and $\phi \Sigma \phi^T = \Sigma$. Then

$$a^T G'_F a = G_F. \quad (18)$$

Thus the Fisher metric is invariant up to the declared perturbation-coordinate reparametrization.

Proof 22. Using $J'a = \phi J$ and $\phi^T \Sigma^{-1} \phi = \Sigma^{-1}$, which follows from $\phi \Sigma \phi^T = \Sigma$, we obtain

$$a^T G'_F a = a^T J'^T \Sigma^{-1} J'a = J^T \phi^T \Sigma^{-1} \phi J = J^T \Sigma^{-1} J = G_F.$$

Corollary 23 (Fisher-only failure mode). Preserving Fisher data is necessary under strict admissibility, but not sufficient for CTMT admissibility. A transformation may preserve G_F while changing N , Σ_N , or resolved-null covariance coupling.

5.2 Null covariance

Let P_N be the projector onto N in a chosen representation. The null covariance block is

$$\Sigma_N = P_N \Sigma P_N. \quad (19)$$

It is part of the full resolution geometry. Null-only covariance changes are therefore genuine defects for strict CTMT and remain defects for conformal CTMT unless they arise from one global positive scale on all of Σ .

5.3 Transport, monodromy, and winding

Definition 24 (Admissible transport). A transport between local charts is admissible if every transition map along the transport path is an automorphism of the relevant resolution geometry.

Proposition 25 (Monodromy as functorial invariant). Let local charts be arranged as a groupoid whose arrows are admissible transports. Resolution geometry defines a functor from this chart groupoid to the groupoid of resolution geometries and their automorphisms. The monodromy of a closed loop is therefore an automorphism of the initial resolution geometry.

Proof 26. Composition of admissible transports is composition of automorphisms, and the identity transport is the identity automorphism. Hence admissible transport respects the groupoid laws. A closed path begins and ends at the same chart, so its induced map lies in the automorphism group of that chart's resolution geometry.

Remark 4 (Tuning-law reinterpretation). Topological or circulation-based tuning laws are best interpreted as diagnostics of transport consistency. They can detect failure of admissible transport, but they are not the primitive definition of morphism admissibility. The primitive definition is preservation of resolution geometry.

6 Automorphism classification theorem

Choose an adapted basis of $\mathcal{O} = R \oplus N$. Write

$$\Sigma = \begin{pmatrix} \Sigma_R & C \\ C^T & \Sigma_N \end{pmatrix}, \quad (20)$$

where Σ_R and Σ_N are positive definite restrictions and $C = C_{RN}$ is the resolved-null cross covariance.

Lemma 27 (Sector preservation gives block form). If $\phi(R) = R$ and $\phi(N) = N$, then in the adapted decomposition

$$\phi = \text{diag}(A, B), \quad A : R \rightarrow R, \quad B : N \rightarrow N. \quad (21)$$

Proof 28. The conditions $\phi(R) = R$ and $\phi(N) = N$ forbid off-diagonal components from R to N and from N to R . Thus ϕ is block diagonal.

Theorem 29 (Strict automorphism group). Let $\mathcal{G} = (\mathcal{O}; R, N, \Sigma)$ with Σ written as in (20). Then

$$\text{Aut}(\mathcal{G}) = \left\{ \text{diag}(A, B) : \begin{array}{l} A\Sigma_R A^T = \Sigma_R, \\ B\Sigma_N B^T = \Sigma_N, \\ ACB^T = C \end{array} \right\}. \quad (22)$$

If $C = 0$, then

$$\text{Aut}(\mathcal{G}) \cong O(\Sigma_R) \times O(\Sigma_N) \cong O(r) \times O(n-r). \quad (23)$$

If $C \neq 0$, the group is the stabilizer of C inside $O(\Sigma_R) \times O(\Sigma_N)$.

Proof 30. By the block-form lemma, $\phi = \text{diag}(A, B)$. Substitution into $\phi\Sigma\phi^T = \Sigma$ gives

$$\begin{pmatrix} A\Sigma_R A^T & ACB^T \\ BC^T A^T & B\Sigma_N B^T \end{pmatrix} = \begin{pmatrix} \Sigma_R & C \\ C^T & \Sigma_N \end{pmatrix},$$

which is exactly (22). When $C = 0$, the two covariance equations decouple. Since every positive definite form is equivalent to the Euclidean form by whitening, $O(\Sigma_R) \cong O(r)$ and $O(\Sigma_N) \cong O(n-r)$.

Theorem 31 (Conformal automorphism group). Let $\mathcal{G}_c = (\mathcal{O}; R, N, [\Sigma])$. Then

$$\text{Aut}(\mathcal{G}_c) = \left\{ \text{diag}(A, B) : \exists \lambda > 0 \text{ such that } \begin{array}{l} A\Sigma_R A^T = \lambda\Sigma_R, \\ B\Sigma_N B^T = \lambda\Sigma_N, \\ ACB^T = \lambda C \end{array} \right\}. \quad (24)$$

If $C = 0$, then

$$\text{Aut}(\mathcal{G}_c) \cong \mathbb{R}_+ \times O(r) \times O(n-r). \quad (25)$$

If $C \neq 0$, the group is the conformal stabilizer of C inside the product conformal group.

Proof 32. The proof is the strict proof with $\phi\Sigma\phi^T = \lambda\Sigma$. If $C = 0$, write $A = \sqrt{\lambda}Q_R$ and $B = \sqrt{\lambda}Q_N$ with $Q_R \in O(\Sigma_R)$ and $Q_N \in O(\Sigma_N)$.

Corollary 33 (Uncoupled Lie dimensions). If $C = 0$, then

$$\dim \text{Aut}(\mathcal{G}) = \frac{r(r-1)}{2} + \frac{(n-r)(n-r-1)}{2}, \quad (26)$$

$$\dim \text{Aut}(\mathcal{G}_c) = 1 + \frac{r(r-1)}{2} + \frac{(n-r)(n-r-1)}{2}. \quad (27)$$

The extra conformal dimension is the global positive covariance scale.

7 Sensor samples to morphism class

The practical computation is as follows.

Step 1. Collect samples. Assemble $y_k \in \mathbb{R}^n$ in a fixed observation basis.

Step 2. Declare transported object. State whether the chart describes a coil response, Navier window, turbine state, diffusion profile, seismic declustered local chart, or another object.

Step 3. Declare admissible perturbations. Choose $\theta \in \mathbb{R}^p$ from protocol-valid perturbations only.

Step 4. Build observation map. Define $F(\theta)$ by measurement, simulation, reconstruction, filtered catalogue statistics, or another declared method.

Step 5. Compute J . Estimate $J = D_\theta F(\theta_0)$ and report rank stability and conditioning.

Step 6. Compute R and N . Use SVD or QR to compute bases for $R = \text{Im } J$ and $N = \text{Ker } J^T$.

Step 7. Estimate covariance. Use sample covariance

$$\hat{\Sigma} = \frac{1}{m-1} \sum_{k=1}^m (y_k - \hat{\mu})(y_k - \hat{\mu})^T$$

or a declared regularized covariance.

Step 8. Block-decompose covariance. In an adapted basis $Q = [Q_R, Q_N]$, compute

$$Q^T \Sigma Q = \begin{pmatrix} \Sigma_R & C \\ C^T & \Sigma_N \end{pmatrix}.$$

Step 9. Classify automorphisms. If $\|C\|/\|\Sigma\|$ is below tolerance, use the block product group. Otherwise compute the stabilizer or conformal stabilizer of C .

Step 10. Test candidate morphisms. For a candidate ϕ , compute

$$\begin{aligned} d_R &= \|P_{\phi R} - P_R\|_F, \\ d_N &= \|P_{\phi N} - P_N\|_F, \\ e_\Sigma &= \frac{\|\phi \Sigma \phi^T - \Sigma\|_F}{\|\Sigma\|_F}, \\ e_{[\Sigma]} &= \min_{\lambda > 0} \frac{\|\phi \Sigma \phi^T - \lambda \Sigma\|_F}{\|\Sigma\|_F}. \end{aligned}$$

Accept strict morphisms when d_R, d_N, e_Σ pass tolerance. Accept conformal morphisms when $d_R, d_N, e_{[\Sigma]}$ pass tolerance.

8 Controlled batteries and operational usefulness

The following batteries are evidence that the axiom is operationally useful, not proof that the axiom is true. They test whether the definition produces stable accept/reject behavior under controlled attacks.

8.1 Strict versus conformal separation

Case	Expected result	d_R	d_N	strict rel.	Σ	conformal residual	Strict	Conformal
identity	pass both	0.0000	0.0000	0.0000		0.0000	PASS	PASS
global covariance scale	fail strict, pass conformal	0.0000	0.0000	3.0000		0.0000	FAIL	PASS
null-only covariance inflation	fail both, not global scale	0.0000	0.0000	1.6762		1.5792	FAIL	FAIL
anisotropic sector rotation	fail strict, generally fail conformal unless conformal	7.47×10^{-16}	6.24×10^{-16}	0.4929		0.4777	FAIL	FAIL
resolved/null mixing	fail both	0.0497	0.0497	0.1160		0.1158	FAIL	FAIL

Empirical finding 1 (Scale valve). The controlled separation battery distinguishes strict and conformal CTMT exactly at global covariance scaling. Null-only covariance inflation is not a global conformal scaling and fails both strict and conformal tests. Resolved/null mixing fails both because it changes the sector decomposition.

8.2 Automorphism groups on tested systems

System	n	r	$n - r$	cross rel.	Strict group	Conformal group
coil axisymmetric clean	24	3	21	3.76×10^{-16}	$O_\Sigma(R) \times O_\Sigma(N)$ $O(3) \times O(21)$	$\cong \mathbb{R}_+ \times O(3) \times O(21)$

System	n	r	$n - r$	cross rel.	Strict group	Conformal group
coil anisotropic instrumented	24	3	21	2.66×10^{-16}	$O_\Sigma(R) \times O_\Sigma(N)$ $O(3) \times O(21)$	$\cong \mathbb{R}_+ \times O(3) \times O(21)$
Navier laminar window	32	8	24	3.34×10^{-16}	$O_\Sigma(R) \times O_\Sigma(N)$ $O(8) \times O(24)$	$\cong \mathbb{R}_+ \times O(8) \times O(24)$
Navier mixed sensor crosscov	32	8	24	0.06699	$\text{Stab}_\Sigma(R, N, C_{RN})$, dim. 120	Lie $\text{CStab}_{[\Sigma]}(R, N, C_{RN})$, Lie dim. 121
diffusion synthetic	28	5	23	3.19×10^{-16}	$O_\Sigma(R) \times O_\Sigma(N)$ $O(5) \times O(23)$	$\cong \mathbb{R}_+ \times O(5) \times O(23)$
reaction-diffusion synthetic	28	5	23	0.06354	$\text{Stab}_\Sigma(R, N, C_{RN})$, dim. 153	Lie $\text{CStab}_{[\Sigma]}(R, N, C_{RN})$, Lie dim. 154
seismic local declustered	20	6	14	0.09259	$\text{Stab}_\Sigma(R, N, C_{RN})$, dim. 58	Lie $\text{CStab}_{[\Sigma]}(R, N, C_{RN})$, Lie dim. 59

Empirical finding 2 (Cross covariance shrinks the group). Systems with negligible resolved-null cross covariance realize the full block-product automorphism group. Systems with non-negligible cross covariance realize a smaller stabilizer group. In the tested classifications, the conformal group has one additional Lie dimension corresponding to global positive scale.

9 Open covariance-scale question

The covariance-scale question remains the main open valve. There are three natural protocol choices.

Strict geometry. Preserve calibrated covariance:

$$\phi \Sigma \phi^T = \Sigma.$$

This is the default when instrument calibration and absolute noise scale are part of the experiment.

Conformal geometry. Preserve covariance up to one global scale:

$$\phi \Sigma \phi^T = \lambda \Sigma, \quad \lambda > 0.$$

This is appropriate only when global covariance scale is explicitly normalized away.

Sector-relative geometry. Preserve only selected covariance structure relative to $R \oplus N$. This could be useful for specialized protocols, but it changes the definition of admissibility and must be tested separately.

Current batteries support a clean distinction between strict and global conformal geometry. They do not justify silently accepting null-only or sector-only covariance rescalings as conformal admissibility.

10 Non-claims and failure modes

Non-claim 34 (No universal physics claim). This paper does not claim a final physical equation, earthquake prediction, Navier–Stokes solution, quantum measurement derivation, or universal laboratory validation. It states and classifies a local morphism-selection structure.

Non-claim 35 (No Fisher-only closure). The result must not be weakened to preservation of G_F alone. Fisher data are induced by resolution geometry and can miss null-sector defects.

Non-claim 36 (No automatic global theorem). Local automorphism classification does not by itself prove global bundle consistency, nonlinear phase-Hessian lift consistency, or holonomy closure across all charts. Those are separate compatibility problems.

Non-claim 37 (No experimental proof of the axiom). The experiments demonstrate stable computation and falsification behavior under the axiom. They do not prove that nature must choose this axiom.

11 Conclusion

The final conservative CTMT morphism statement is:

$$\phi \in \text{Mor}_{\text{CTMT}} \iff \phi^* \mathcal{G} = \mathcal{G}, \quad \mathcal{G} = (\mathcal{O}; R, N, \Sigma).$$

The notation (R, N, Σ) is a representation of a coordinate-free object: the decomposition of observation space into resolved and unresolved subspaces together with the covariance structure induced by the measurement protocol.

Under this axiom, rank preservation, sector preservation, covariance compatibility, Fisher invariance, null-sector diagnostics, and monodromy compatibility become consequences rather than primitive assumptions. The automorphism group is completely determined by the block covariance structure in the resolved/null decomposition. If resolved–null cross covariance vanishes, the group is the block orthogonal product. If cross covariance is present, the group shrinks to a stabilizer. If global covariance scale is quotiented out by protocol, exactly one conformal scale dimension is added.

Thus CTMT is no longer formulated as a path from tuning law to Fisher metric to geometry. It is formulated as a direct operational geometry:

$$\text{samples} \rightarrow J \rightarrow (R, N, \Sigma) \rightarrow \text{Aut}(R, N, \Sigma) \rightarrow \text{diagnostics}.$$

The strongest honest conclusion is that the local morphism problem has stabilized into a precise geometry with computable automorphism classes. The remaining open work is to decide the covariance-scale protocol, test sector-relative alternatives if needed, and extend local automorphism compatibility to nonlinear and global transport settings.